PRNU: The Robustness of Source Camera Identification against Compression

Chinasa Njoku
Dept. of Artificial Intelligence and Human Interfaces
University of Salzburg (PLUS)
J.-Haringerstr.2, 5020 Salzburg, Austria
njokuchinasa@gmail.com

***Abstract*—With the continued growth and adoption of digital media, the need to attribute digital images to their sources after they have been acquired remains a fundamental problem in multimedia forensics. So far, many studies have identified that Source Camera Identification (SCI) using Photo Response Non-Uniformity (PRNU) degrades under compression. These studies mostly assess baseline JPEG and its impact at different Quality Factors (QFs). However, for a given QF, different images compress to different sizes. This is problematic for evaluating the effect of compression in the real-world setting, as in most networks and storage systems the target size is often the fixed parameter while the QF varies per image. In this paper we evaluate the SCI performace under compression at different target bitrates, simulating real-world strict bandwidth and storage constraints. We also extend this evaluation to the more recent compression schemes like JPEG 2000, JPEG XR, JPEG XL, JPEG AI and CompressAI. This is to provides a codec recommendation for severe image degradation at very low bitrates that mostly preserve the integrity of the forensic signal required for SCI. For the experiment, the Adobe Fivek dataset was used with 17 cameras selected between three different camera brands (NIKON, Canon and Sony).**

***Index Terms*—Photo Response Non-Uniformity (PRNU), Source Camera Identification (SCI), image compression, image forensics, autoencoders, image quality metrics.**

I. Introduction

Once an image is acquired and shared, it becomes quite difficult to trace back to the originating sensor if it is not explicitly labeled. Over the years, several methods have been developed for reliable SCI. A very popular technique relies on the unique sensor artifacts that are present in the image scene. This is also know as PRNU.

The PRNU is formed as a result slight microscopic variations in the pixel silicon during manufacturing that causes individual pixels to have different sensitivities to light. Therefore once two or more different cameras are used to capture a particular scene, we can reliably trace which camera generated what image from the noise residue in each image if we know the PRNU reference pattern of the cameras.

This method of SCI is highly effective. But in real world applications the acquired images are rarely in their raw state during source attribution. In other to store or transfer them over networks, these images undergo compression most times. Many studies already show that SCI performace degrades as more scene content is removed during compression. With baseline JPEG, as images are comppressed varying the QFs. However, different images compress to different sizes given the same QF. This is particularly challenging if we properly evaluate the imapact of compression under real-world constraints as most network and storage systems are designed to vary QFs in other to achieve a target bitrate or size.

In this work, we are going to

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Number equations consecutively. To make your equations more compact, you may use the solidus (/), the exp function, or appropriate exponents. Italicize Roman symbols for quantities and variables, but not Greek symbols. Use a long dash rather than a hyphen for a minus sign. Punctuate equations with commas or periods when they are part of a sentence, as in:

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TABLE I
TABLE TYPE STYLES

Table Head	**Table Column Head**		
	Table column subhead	***Subhead***	***Subhead***
copy	More table copy[a]		

[a]Sample of a Table footnote.

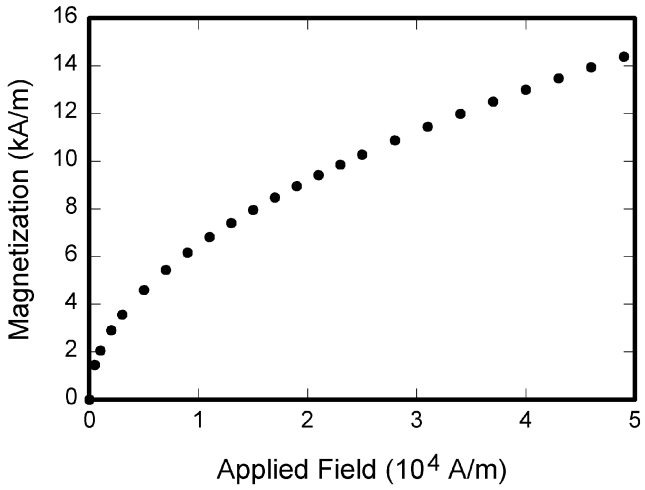


Fig. 1. Example of a figure caption.

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ACKNOWLEDGMENT

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Number footnotes separately in superscripts. Place the actual footnote at the bottom of the column in which it was cited. Do not put footnotes in the abstract or reference list. Use letters for table footnotes.

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